Infinite geometric progressions and the binomial expansion [Cambridge revision]

A sum that never ends but still has a value, and a triangle that expands any bracket.

LESSON 101–102 | 2 × 40 MIN | REVISION OF GRADE 9, §22 | P1 7.4, 6.1–6.3

LESSON CLOCK

20' | 20' | 25' | 25' | 15' | 5

When an infinite sum exists	20 min
Recurring decimals	20 min
Pascal's triangle	25 min
The binomial formula	25 min
One term without the rest	15 min
Homework	5 min

LEARNING OBJECTIVES

- State when an infinite geometric series converges, and find its sum.
- Convert a recurring decimal to a fraction with the sum to infinity.
- Expand $(a + b)^n$ using Pascal's triangle and the binomial coefficients.
- Find one named term of an expansion without writing the others.

TEXTBOOK

National	Algebra 9, pp. 215–228
Cambridge	Pure Mathematics 1, pp. 128–147, 163–168
Workbook	Exercise 6A–6C, 7D

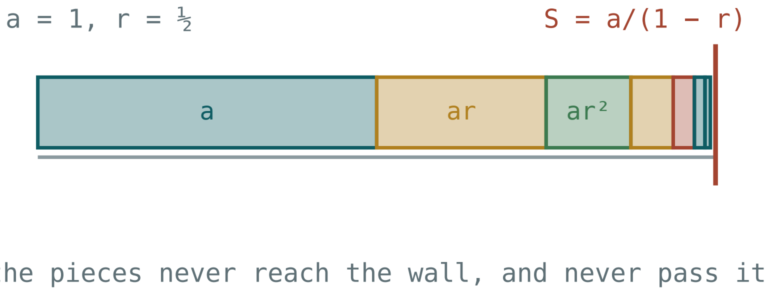
01 Explanation

When an infinite sum exists

Add $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \ldots$ and the total never passes 2. Add $1 + 2 + 4 + 8 + \ldots$ and it passes everything. The difference is the ratio.

$$S\_\infty = \frac{a}{1 - r}, \text{ valid exactly when } |r| < 1$$

Why: in $S_n = \frac{a(1 - r^n)}{1 - r}$, the term r^n shrinks to zero when $|r| < 1$ and grows without limit otherwise.



Each piece is half the last — the pieces approach the wall and never reach it.

> $|R| < 1$ IS THE WHOLE CONDITION
>
> $r = 1$ gives $a + a + a + \ldots$, which diverges; $r = -1$ gives $a - a + a - \ldots$, which oscillates and has no sum. Both are excluded, and a question that asks “for what x does the series converge” is asking you to solve $|r| < 1$.

Recurring decimals

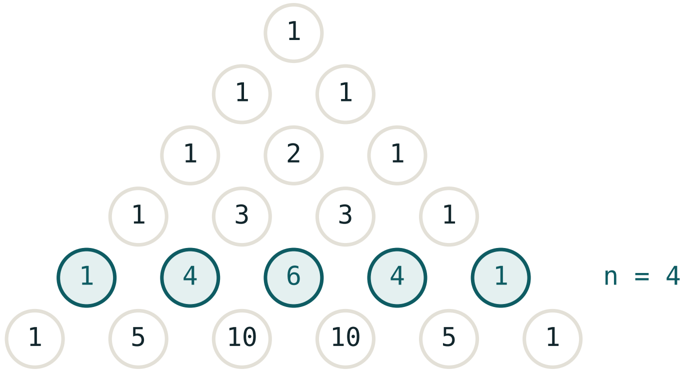
Every recurring decimal is a geometric series in disguise, so every one is a fraction.

DECIMAL	AS A SERIES	A	R	FRACTION
$0.333\ldots$	$0.3 + 0.03 + \ldots$	0.3	0.1	$\frac{1}{3}$
$0.272727\ldots$	$0.27 + 0.0027 + \ldots$	0.27	0.01	$\frac{3}{11}$
$0.999\ldots$	$0.9 + 0.09 + \ldots$	0.9	0.1	1

The last line is not a trick: $\frac{0.9}{1 - 0.1} = \frac{0.9}{0.9} = 1$. The decimal $0.999\ldots$ **is** 1, written differently.

Pascal’s triangle

Expand $(a + b)^n$ for small n and the coefficients form a triangle in which every entry is the sum of the two above it.



each entry is the sum of the two above it

Row n gives the coefficients of $(a + b)^n$; the row highlighted is n = 4.

N	COEFFICIENTS	EXPANSION
2	1 2 1	$a^2 + 2ab + b^2$
3	1 3 3 1	$a^3 + 3a^2b + 3ab^2 + b^3$
4	1 4 6 4 1	$a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$
5	1 5 10 10 5 1	$a^5 + 5a^4b + \ldots + b^5$

TWO PATTERNS TO CHECK EVERY EXPANSION AGAINST

The powers of a fall from n to 0 while those of b rise from 0 to n, so **every term's powers add to n**. And there are always $n + 1$ terms.

The binomial formula

$$(a + b)^n = \Sigma_{k=0}^{n} C(n, k)\, a^{(n-k)}\, b^{k}$$

$$C(n, k) = \frac{n!}{k!(n-k)!}$$

The coefficient $C(n, k)$ — written $^{n}C_k$ or $(^{n}_{k})$ — is exactly the entry in row n, position k, of Pascal's triangle. For n up to about 6 the triangle is faster; beyond that, the factorial formula is.

$C(N, K)$	VALUE	WHY
$C(5, 0)$	1	one way to choose nothing
$C(5, 1)$	5	$= n$
$C(5, 2)$	10	$\frac{5 \times 4}{2}$
$C(5, 3)$	10	$= C(5, 2)$
$C(8, 3)$	56	$\frac{8 \times 7 \times 6}{3 \times 2 \times 1}$

THE SYMMETRY THAT HALVES THE WORK

$$C(n, k) = C(n, n - k)$$

Choosing k to include is the same as choosing $n - k$ to leave out. So $C(20, 18) = C(20, 2) = 190$ — three multiplications instead of eighteen.

One term without the rest

Examinations rarely want the whole expansion. They want one coefficient.

> the term in b^k is $C(n, k)\, a^{(n-k)}\, b^k$

Example. The coefficient of x^3 in $(2 + x)^5$: here $k = 3$, $a = 2$, so the term is $C(5, 3)·2^2·x^3 = 10 \times 4\, x^3 = 40x^3$.

With a coefficient on x, raise it too. In $(1 + 3x)^4$, the term in x^2 is $C(4, 2)(3x)^2 = 6 \times 9x^2 = 54x^2$ — the 3 is squared as well.

A MINUS SIGN BELONGS TO THE WHOLE BRACKET

In $(1 - 2x)^5$ write $b = -2x$, so the term in x^3 is $C(5, 3)(-2x)^3 = 10(-8x^3) = -80x^3$. Forgetting to cube the minus is the single commonest error in this topic.

02 Worked examples

EXAMPLE 1 **Find the sum to infinity of $12 + 4 + \frac{4}{3} + \dots$.**

1. $r = \frac{4}{12} = \frac{1}{3}$

   $|r| < 1$ — it converges.

2. $S_\infty = \dfrac{12}{1 - \frac{1}{3}}$

3. $= \dfrac{12}{\frac{2}{3}} = 18$

ANSWER

18

EXAMPLE 2 **Write $0.272727\dots$ as a fraction.**

1. $a = 0.27$, $r = 0.01$

2. $S = \frac{0.27}{0.99}$

3. $= \frac{27}{99} = \frac{3}{11}$

ANSWER

$\frac{3}{11}$

EXAMPLE 3 **Expand $(2 + x)^4$ fully.**

1. Row 4: $1, 4, 6, 4, 1$.
2. $2^4 + 4(2^3)x + 6(2^2)x^2 + 4(2)x^3 + x^4$
3. $16 + 32x + 24x^2 + 8x^3 + x^4$

   Five terms, powers adding to 4.

ANSWER

$16 + 32x + 24x^2 + 8x^3 + x^4$

EXAMPLE 4 **Find the coefficient of x^3 in $(1 - 2x)^6$.**

1. $C(6, 3) = 20$
2. $(-2x)^3 = -8x^3$

   Cube the whole bracket.
3. $20 \times (-8) = -160$

ANSWER

-160

03 Interactive model

Cut a strip of paper in half repeatedly and lay the pieces end to end — the sum to infinity appears as a finite length.

Convergence, and coefficients

S_∞ exists when:

- $r > 0$
- $|r| < 1$
- $r < 1$
- always

S_∞ of $1 + \frac{1}{2} + \frac{1}{4} + \dots$:

- 1
- 2
- ∞
- $\frac{3}{2}$

$0.999\dots$ equals:

- 0.99
- 1
- just under 1
- $\frac{9}{10}$

Row 4 of Pascal's triangle:

- $1\ 3\ 3\ 1$
- $1\ 4\ 6\ 4\ 1$
- $1\ 5\ 10\ 10\ 5\ 1$
- $1\ 4\ 4\ 1$

$(a + b)^5$ has how many terms?

- 5
- 6
- 7
- 10

$C(8, 3)$ is:

- 24
- 56
- 336
- 112

$C(20, 18)$ equals:

- $C(20, 2)$
- $C(20, 18)$ only
- 20
- 180

The term in x^3 of $(1 - 2x)^6$:

- $160x^3$
- $-160x^3$
- $-20x^3$
- $-8x^3$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Infinite series	Cheksiz qator	Бесконечный ряд
Sum to infinity	Cheksiz yig‘indi	Сумма бесконечной прогрессии
Convergent	Yaqinlashuvchi	Сходящийся
Divergent	Uzoqlashuvchi	Расходящийся
Recurring decimal	Davriy o‘nli kasr	Периодическая дробь
Binomial expansion	Binomial yoyilma	Разложение бинома
Binomial coefficient	Binomial koeffitsiyent	Биномиальный коэффициент
Pascal’s triangle	Paskal uchburchagi	Треугольник Паскаля
Factorial	Faktorial	Факториал
General term	Umumiy had	Общий член

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

An infinite GP converges when:

- $|r| < 1$
- $r > 1$
- $a > 0$
- always

S_∞ equals:

- $\frac{a}{1 + r}$
- $\frac{a}{1 - r}$
- $\frac{a}{r}$
- ar

Every entry of Pascal's triangle is:

- a prime
- the sum of the two above
- a square
- random

In every term of $(a + b)^n$ the powers:

- add to n
- are equal
- add to $2n$
- add to k

$C(n, k) = C(n, n - k)$ because:

- of the formula only
- choosing k in is choosing $n - k$ out
- it is a coincidence
- of Pascal

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. S_∞ of $1 + \frac{1}{2} + \frac{1}{4} + \ldots$

   ANSWER 2

2. S_∞ of $9 + 3 + 1 + \ldots$

   ANSWER 13.5

3. Does $1 + 2 + 4 + \ldots$ converge?

   ANSWER No — $r = 2$

4. $0.333\ldots$ as a fraction

   ANSWER $\frac{1}{3}$

5. Row 3 of Pascal's triangle

   ANSWER $1\ 3\ 3\ 1$

6. Number of terms in $(a + b)^7$

   ANSWER 8

7. $C(6, 2)$

   ANSWER 15

Medium · the standard question

7 PROBLEMS

1. S_∞ of $12 + 4 + \frac{4}{3} + \dots$

   ANSWER 18

2. S_∞ of $8 - 4 + 2 - \dots$

   ANSWER $\frac{16}{3}$

3. $0.272727\dots$ as a fraction

   ANSWER $\frac{3}{11}$

4. Expand $(1 + x)^4$

   ANSWER $1 + 4x + 6x^2 + 4x^3 + x^4$

5. Expand $(2 + x)^4$

   ANSWER $16 + 32x + 24x^2 + 8x^3 + x^4$

6. Coefficient of x^2 in $(1 + 3x)^4$

   ANSWER 54

7. $C(8, 3)$

   ANSWER 56

Hard · stretch and reason

7 PROBLEMS

1. Coefficient of x^3 in $(1 - 2x)^6$

   ANSWER -160

2. Coefficient of x^4 in $(2 - x)^7$

   ANSWER 280

3. Constant term in $(x + \frac{2}{x})^6$

   ANSWER 160

4. For which x does $1 + 2x + 4x^2 + \ldots$ converge, and to what?

   ANSWER $|x| < \frac{1}{2}$; $\frac{1}{1-2x}$

5. A GP has $S\_\infty = 27$ and $a = 18$: find r

   ANSWER $r = \frac{1}{3}$

6. A ball dropped from 10 m rebounds 0.6 of its height each time: total distance travelled

   ANSWER 40 m

7. Show that $C(n, k) + C(n, k + 1) = C(n + 1, k + 1)$

   ANSWER Pascal's rule, from the factorial form

07 Homework

Homework — 5 tasks

This is the last algebra lesson of Grade 10; keep the summary page for Grade 11.

1. Find the sum to infinity of $20 + 5 + 1.25 + \ldots$, and state the value of r.
2. Write $0.4545\ldots$ and $0.1666\ldots$ as fractions using the sum to infinity.
3. Expand $(3 - x)^4$ fully, and check that the powers in each term add to 4.
4. Find the coefficient of x^3 in $(2 + 3x)^5$ without writing the other terms.

5. A GP has sum to infinity 12 and first term 8. Find r and the fourth term.